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OPERATING SYSTEMS LABORATORY

Course Code: CSE309R01

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Lab Manual

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Exercise No. 3 IPC using Message Queue

Write program to carry out inter-process communication using message

queue Objectives

To implement IPC using message queues with the help of the library functions

available **Prerequisite**

Knowledge of IPC, Message queues, syntax and functionalities

Theory

- A **message queue** is an inter-process communication (IPC) mechanism that allows processes to exchange data in the form of messages between two processes • It allows processes to communicate asynchronously by sending messages to each other where the messages are stored in a queue, waiting to be processed, and are deleted after being processed.

Procedure

- ☐ Create the sender process and receiver process
- ☐ Create a message queue using the appropriate functions
- ☐ Sender pushes its message into message queue using msgsnd
- ☐ Receiver retrieves the message using msgrcv and show it to the user

Sample Input

1 400

Y

2 1798

Y

3 1500

N

Output

```
guest-m22c35@cnlab:~$ gcc server.c
guest-m22c35@cnlab:~$ ./a.out
Enter no & amnt 4 msg Q : 1 400
One more need (y/n) : y
Enter no & amnt 4 msg Q : 2 1798
One more need (y/n) : y
Enter no & amnt 4 msg Q : 3 1500
One more need (y/n) : n
guest-m22c35@cnlab:~$

File Edit View Search Terminal Help
guest-m22c35@cnlab:~$ gcc client.c
guest-m22c35@cnlab:~$ ./a.out
Enter the msg type : 1

The client of no : 1
Should pay : 400.000000
One more needed (y/n) : y
Enter the msg type : 3

The client of no : 3
Should pay : 1500.000000
One more needed (y/n) : y\
Enter the msg type : 2

The client of no : 2
Should pay : 1798.000000
```

Observation

Summary of Message Queue System Calls: Write the syntax of the following in detail:

1. **msgget():** Creates or accesses a message queue.

Syntax: `int msgget(key_t key, int msgflg);`

Creates a new message queue or accesses an existing message queue and returns its ID.

2. **msgsnd():** Sends a message to the message queue.

Syntax: `int msgsnd(int msqid, const void *msgp, size_t msgsz, int msgflg);`

Sends a message from the sender process to the specified message queue.

3. **msgrcv():** Receives a message from the message queue.

Syntax: `ssize_t msgrcv(int msqid, void *msgp, size_t msgsz, long msgtyp, int msgflg);`

Receives a message from the specified message queue into the receiver process.

4. **msgctl():** Controls the message queue (e.g., retrieves status, removes the queue, changes permissions).

Syntax: `int msgctl(int msqid, int cmd, struct msqid_ds *buf);`

Performs operations such as retrieving queue information, changing permissions, or deleting the message queue.

Result

Thus, inter-process communication using Message Queue was implemented successfully, and the sender and receiver processes exchanged data correctly.